



The mathematics of electorate allocation in New Zealand based on the outcome of the 2013 Census and Māori Electoral Option 2013



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1 Purpose and background

Purpose

This paper outlines how the number of electorates in New Zealand is calculated, as specified in the Electoral Act 1993 (available from www.legislation.govt.nz). It describes how concepts from the Act were implemented in the Government Statistician's electoral population calculations, using the results of the 2013 Census of Population and Dwellings and the Māori Electoral Option 2013.

Background to electoral population and electorate calculations

The Electoral Act 1993 arose from an overhaul of the New Zealand electoral system, with major changes being made to the representation of the Māori population. One of the major changes under the Act was to make the number of Māori electorates vary according to the proportion of Māori who choose to be on the separate Māori electoral roll. The number of general electorates was already variable and this ensured consistency. The allocation of electorates to the two electoral populations – Māori and general – is effectively based on four concepts.

Concept 1

Electoral representation is based on the census usually resident population count in New Zealand, including children and others not eligible to vote. This is sourced from the Census of Population and Dwellings, which is normally held every five years (Electoral Act 1993, s3(1)).

Concept 2

The census usually resident population count is split into two segments:

- the Māori electoral population (MEP) – a proportion of the electoral Māori descent census usually resident population count. This is based on the percentage of Māori descent electors who choose the Māori electoral roll (this includes electors who are registered on the main Māori electoral roll or the dormant Māori electoral roll). Refer to [Number of Electorates and Electoral Populations: 2013 Census](#) (Statistics NZ, 2013a) for a description of how Māori descent is calculated in the census for electoral purposes.
- the general electoral population (GEP) – the electoral non-Māori descent census usually resident population count, and a proportion of the electoral Māori descent census usually resident population count who choose the general electoral roll (this includes electors who are registered on the main general electoral roll or the dormant general electoral roll) (Electoral Act 1993, s3(1)).

Concept 3

There are approximately equal electoral populations in each Māori and general electorate (Electoral Act 1993, s35(3) and s45(3)).

Concept 4

There are 16 general electorates in the South Island. (Electoral Act 1993, s35(3a)).

The following sections outline the mathematics for the allocation of electorates. The mathematics have not changed since 2007 (Statistics NZ, 2013b).

2 Electoral population calculations

Sections 35(3b, 3c) and 45(3a) of the Electoral Act 1993 specify that the calculations of electoral populations be carried out separately for New Zealand and the North and South Islands.

To determine the number of electorates, the Act divides the census usually resident population count of New Zealand into three groups:

- the North Island general electoral population (NI GEP)
- the South Island general electoral population (SI GEP)
- the Māori electoral population (MEP).

The methods for calculating the three electoral populations are provided below.

Māori electoral population

See section 1, concept 2.

We define the Māori electoral population (MEP) as the proportion of enrolled Māori descent electors who choose the Māori electoral roll, multiplied by the electoral Māori descent census usually resident population count from the 2013 Census. This means that the MEP includes people who are not enrolled on the electoral roll (such as children). The actual calculation uses the following formula:

$$MEP = \frac{m}{m + g} d$$

where:

m is the total number of Māori descent electors registered on the Māori electoral roll,

g is the total number of Māori descent electors registered on the general electoral roll, and,

d is the total number of people with Māori descent in the census.

The proportion of enrolled Māori descent electors who choose the Māori electoral roll is of critical importance in all these calculations. We will refer to it as the Māori ratio (r):

$$r = \frac{m}{m + g}$$

The MEP is calculated at the national level using national totals, see table 2.1.

Table 2.1

Calculation of 2013 Māori electoral population of New Zealand

Population	Source	Number
Census usually resident population count (u)	2013 Census	4,242,049
Electoral Māori descent census usually resident population count (d)	2013 Census	755,598
Māori on Māori electoral roll (m)	Māori electoral roll	256,212
Māori on general electoral roll (g)	General electoral roll	203,640
% of Māori choosing Māori electoral roll (Māori ratio, r)	$r = m/(m + g)$	55.7
Māori electoral population	$MEP = r \times d$	420,990
Note: The New Zealand MEP is 420,990 (not 420,068), as it is calculated using the Māori ratio of New Zealand rather than the Māori ratios for the North and South Islands.		

General electoral population

See section 1, concept 2.

The general electoral population (GEP) is the census usually resident population count (u) less the Māori electoral population.

$$GEP = u - MEP$$

To calculate the number of general electorates we need to calculate the GEP for each island, see table 2.2. The MEPs calculated here, at the island level, are only used for calculating the NI GEP and the SI GEP.

Table 2.2

Calculation of 2013 general electoral populations for each island

Population	North Island	South Island	Chatham Islands
Census usually resident population count (u)	3,237,049	1,004,400	600
Electoral Māori descent census usually resident population count (d)	643,576	111,643	379
Māori on Māori electoral roll (m)	227,661	28,403	148
Māori on general electoral roll (g)	168,397	35,153	90
% of Māori choosing Māori electoral roll (r)	57.5	44.7	62.2
Māori electoral population (MEP) ($r \times d$)	369,939	49,893	236
General electoral population (GEP) ($u - MEP$)	3,237,049 - 369,939 = 2,867,110	1,004,400 - 49,893 = 954,507	600 - 236 = 364

The Electoral Act 1993 divides the census usually resident population count into three groups as stated previously. Section 46(2) of the Act specifies how the GEP and MEP for the Chatham Islands will be treated. To complete the calculation of the GEPs (NI and SI), the Chatham Islands GEP is added to the South Island, while the Chatham Islands MEP is added to the North Island, shown in table 2.3.

Table 2.3

Calculation of 2013 general electoral populations in the North and South Islands

Population	North Island	South Island
Total electoral population	3,237,049 +236 = 3,237,285	1,004,400 +364 = 1,004,764
Māori electoral population	369,939 +236 = 370,175	49,893
General electoral population	3,237,285 - 370,175 NI GEP = 2,867,110	1,004,764 - 49,893 SI GEP = 954,871

3 Electorate calculations

South Island electorates and quota

See section 1, concept 4.

The number of general electorates in the South Island is fixed at 16. Therefore, dividing the SI GEP by 16 gives the South Island quota (q_s), the average size of a South Island general electorate.

For the 2013 calculations:

$$\begin{aligned}
 q_s &= 954,871 / 16 \\
 &= 59,679
 \end{aligned}$$

Māori electorates and quota

See section 1, concept 3.

The populations for each electorate should be approximately equal. Therefore, the South Island quota is used to determine the number of Māori and general electorates.

In the 2013 calculations, the number of Māori electorates (n_m) is determined by:

$$\begin{aligned}
 n_m &= \text{MEP} / q_s \\
 &= 420,990 / 59,679 \\
 &= 7.05
 \end{aligned}$$

This is rounded to the nearest whole number to give seven Māori electorates in 2013.

The Māori quota (q_m), or average population for Māori electorates, is:

$$\begin{aligned}
 q_m &= \text{MEP} / n_m \\
 &= 420,990 / 7 \\
 &= 60,141
 \end{aligned}$$

North Island electorates and quota

The South Island quota is used to determine the number of North Island electorates.

In the 2013 calculations, the number of North Island electorates (n_n) is determined by:

$$\begin{aligned} n_n &= \text{NI GEP} / q_s \\ &= 2,867,110 / 59,679 \\ &= 48.04 \end{aligned}$$

This is rounded to the nearest whole number to give 48 North Island electorates. The quota for North Island electorates (q_n) is:

$$\begin{aligned} q_n &= \text{NI GEP} / n_n \\ &= 2,867,110 / 48 \\ &= 59,731 \end{aligned}$$

Approximately equal electoral populations

The method for calculating the number of electorates ensures that the individual quotas are as near to equal as possible, given that the number of electorates in each case has to be rounded to a whole number.

Sections 36 and 45(7) of the Electoral Act 1993 state that electorate populations may vary by up to 5 percent of their quota, shown in table 3.1.

Table 3.1

Summary of 2013 electoral calculations

Population	Māori	North Island general	South Island general
Electoral population	420,990	2,867,110	954,871
Number of electorates	7	48	16
Quota	60,141	59,731	59,679
Minimum electoral population	57,134	56,745	56,696
Maximum electoral population	63,148	62,717	62,662

4 Setting new boundaries

Once the Government Statistician has carried out these calculations and reported them to the Surveyor General and other members of the Representation Commission, it is the task of the Surveyor General to draw up provisional boundaries. The Commission releases the proposed names and boundaries for public consultation and will hear objections and counter-objections before finalising the new electorate boundaries in April 2014.



References and further reading

References

[Electoral Act 1993](#). Available from www.legislation.govt.nz.

Statistics NZ (2013a). [Number of Electorates and Electoral Populations: 2013 Census](#). Available from www.stats.govt.nz.

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Further reading

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